CHRONIC KIDNEY DISEASE PREDICTION USING LOGISTIC REGRESSION

AI-101 MINI PROJECT · GROUP-26

GITHUB REPO

DOCUMENTATION LEAD

Suda Tanishq Reddy
2026MCB1573

EXECUTION LEAD

Abhishek
2026DAB1290

PRESENTATION LEAD

Prabhnoor Kataria
2026AIB1066

VIDEO LEAD

Davuluri Vinod
2026EEB1339

PREDICTIVE HEALTHCARE · MACHINE LEARNING · BINARY CLASSIFICATION

THE PROBLEM AND DATASET

THE PROBLEM

- Chronic Kidney Disease (CKD) often develops gradually, making early identification challenging.
- Analyzing patient health indicators can help identify potential CKD cases for timely medical evaluation.
- We aim to use Machine Learning to predict whether a patient has CKD.

DATASET OVERVIEW

Dataset:	Chronic Kidney Disease Dataset
Dataset source:	Chronic_Kidney_Disease_data.csv
Records:	**1,659 patients**
Features:	**54 columns**
Target Variable:	Diagnosis (CKD / No CKD)
Algorithm:	Logistic Regression

MODEL ARCHITECTURE & TECHNOLOGIES USED

INPUT DATA

- Chronic Kidney Disease dataset
- 1,659 patient records
- 54 columns
- Target variable: Diagnosis

DATA SPLITTING

- Train-test split
- 80% Training Data
- 20% Testing Data
- Stratified sampling

DATA PREPROCESSING

Column Transformer

NUMERICAL FEATURES

- Median Imputation
- Standard Scaling

CATEGORICAL FEATURES

- Mode Imputation
- One-Hot coding

CLASSIFICATION MODEL

- Logistic Regression
- Scikit-learn
- Solver: liblinear
- Maximum iterations: 12,828

OUTPUT

- Predicted Diagnosis
- Binary Classification
- Class 0 / Class 1

Training: model learns from training data

Prediction: trained model generates predictions for testing data

LIBRARIES USED

Pandas

Load and manipulate data

NumPy

Numerical and array operations

Matplotlib

Plots and visualization

Seaborn

Statistical plots

Scikit-learn

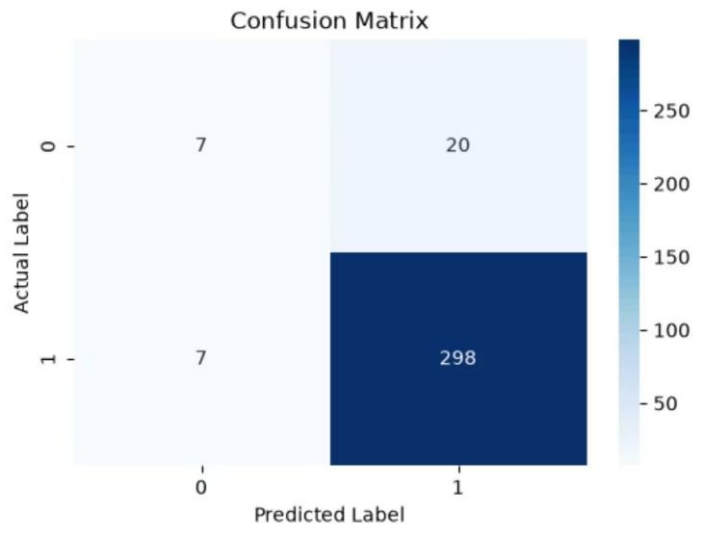
Preprocess, train and evaluate

MODEL PERFORMANCE & EVALUATION

ACCURACY	PRECISION	RECALL	F1-SCORE
91.9%	94.0%	97.4%	95.7%

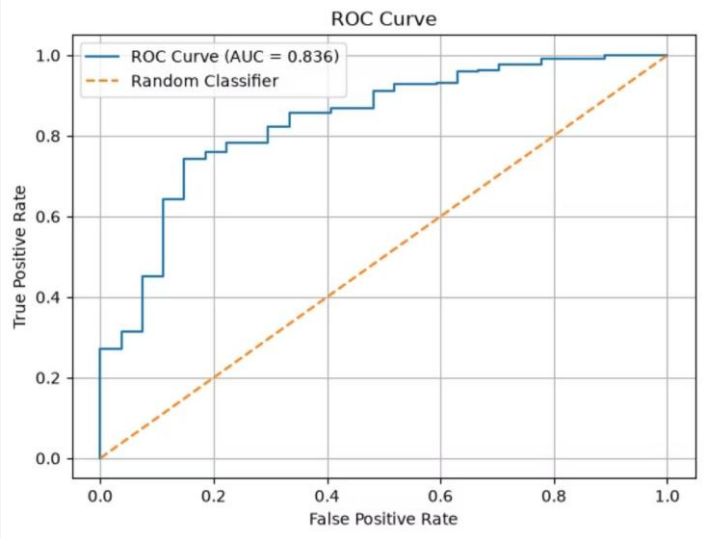
CONFUSION MATRIX

·ACTUAL × PREDICTED DIAGNOSIS



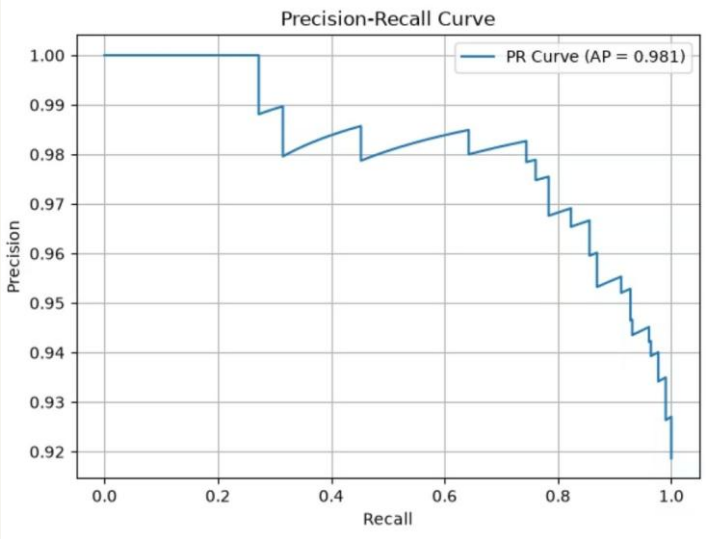
ROC CURVE

·ROC-AUC = **0.836**



PRECISION–RECALL CURVE

·AVERAGE PRECISION = **0.981**



KEY OBSERVATIONS

- The model achieves 91.9% accuracy and demonstrates strong CKD detection, but struggles to identify non-CKD cases (26% recall).
- This highlights the impact of class imbalance in the test dataset (305 CKD vs 27 non-CKD cases).

PROJECT LEARNINGS & TEAM CONTRIBUTIONS

Chronic Kidney Disease Prediction using Logistic Regression.

WHAT WORKED

- Scikit-learn preprocessing pipelines
- Missing-value handling and feature scaling
- Multiple evaluation metrics

WHAT DIDN'T WORK AS EXPECTED

- Class imbalance affected minority-class detection
- Correlation analysis showed limited scope for reducing features; `DoctorInCharge` had low predictive relevance but remained the only categorical feature.

SUDA TANISHQ REDDY

DOCUMENTATION LEAD

- Dataset exploration
- README architecture
- Notebook integration
- Github Repo

ABHISHEK

EXECUTION LEAD

- Conda environment
- Dependency management
- Preprocessing

PRABHNOOR KATARIA

PPT LEAD

- Logistic Regression architecture
- Model training
- Presentation

DAVULURI VINOD

VIDEO LEAD

- Model evaluation
- ROC/PR curves
- Metrics analysis
- Video demo

AI TOOLS WERE USED FOR UNDERSTANDING PYTHON, PANDAS AND SCIKIT-LEARN, DEBUGGING CODE, AND PREPARING PROJECT DOCUMENTATION AND PRESENTATIONS.